

Pendulum Phase Transport — KN64

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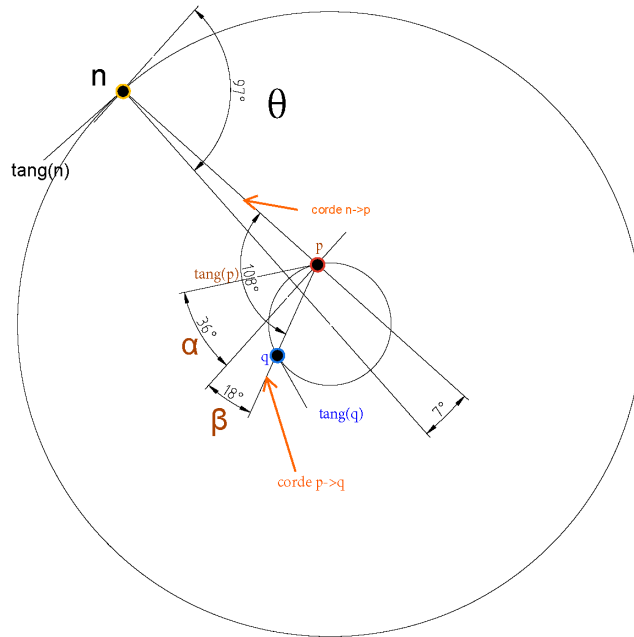
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Structural asymmetry of α and conditional localization of β

This general study follows the analysis of the dyadic structure of prime number distribution initiated in December 2025. It opens a path to postulate the encoding of p in n and, in a pendular effect, of q in p , as schematically postulated below.

Conventions et définitions

- > n est le semiprime issu du produit $p \cdot q$
- > p est le plus petit facteur premier
- > q est le plus grand des facteurs premiers
- > l'angle theta θ est calculé à partir du rayon de n et de la $\perp$ à la corde $n \rightarrow p$
- > l'angle alpha α est calculé à partir de la tangente de $\text{tang}(p)$ et de la $\perp$ à la corde $n \rightarrow p$
- > l'angle Beta β est constitué en symétrie (parfois complémentaire) de la $\perp$ à la corde $n \rightarrow p$ de la $\perp$ à la corde $n \rightarrow q$



1. Context and Objective

This note formalizes an advancement derived from the analysis of the K64 dataset, representing a set of data related to triplets (n, p, q) , where n is a semiprime product of two prime factors p and q . This data is compiled in the file:

pendulum_triplets_64_500k_with_beta.jsonl

The command to generate the statistics and figures supporting this work is:

```
python .\phase_transport\tools\analyze_pendulum_angles.py `
>> --jsonl phase_transport\outputs\pendulum\pendulum_triplets_64_500k_with_beta.jsonl `
>> --tag KN64 `
>> --out phase_transport\outputs `
>> --bins 181 `
>> --alpha_lobe_mode pi_lobes `
>> --near_pi_eps 0.05
```

This dataset is generated from **500,000 triplets (n, p, q)** of 64-bit semiprimes and analyzed using:

analyze_pendulum_angles.py

The objective is to study the evolution of the **joint angular structure** of the following variables, as we interpret the relationships between various quadrants:

- θ_n : angle associated with n (semiprime)
- α : angle associated with p
- β : angle associated with q

within the framework of the **pendular/concentric model** at the origin of the project.

2. Angular Convention

All angular variables are:

- Expressed in **radians**
- Folded into the interval $(-\pi, \pi]$
- Oriented **counterclockwise (trigonometric direction)**

3. Reference Figures (KN64)

All figures below are automatically generated and stored in:

`phase_transport/outputs/figures/KN64/`

Marginal Distributions

- *Histogram of θ_n (density):*
`hist_theta_n_density_KN64__20260202_120748.png`
- *Histogram of α (density):*
`hist_alpha_density_KN64__20260202_120748.png`
- *Histogram of β (density):*
`hist_beta_density_KN64__20260202_120748.png`

Conditional Distributions

- β | α in the positif lobe
`hist_beta_given_alpha_pos_density_KN64__20260202_120748.png`
- β | α in a negative lobe
`hist_beta_given_alpha_neg_density_KN64__20260202_120748.png`

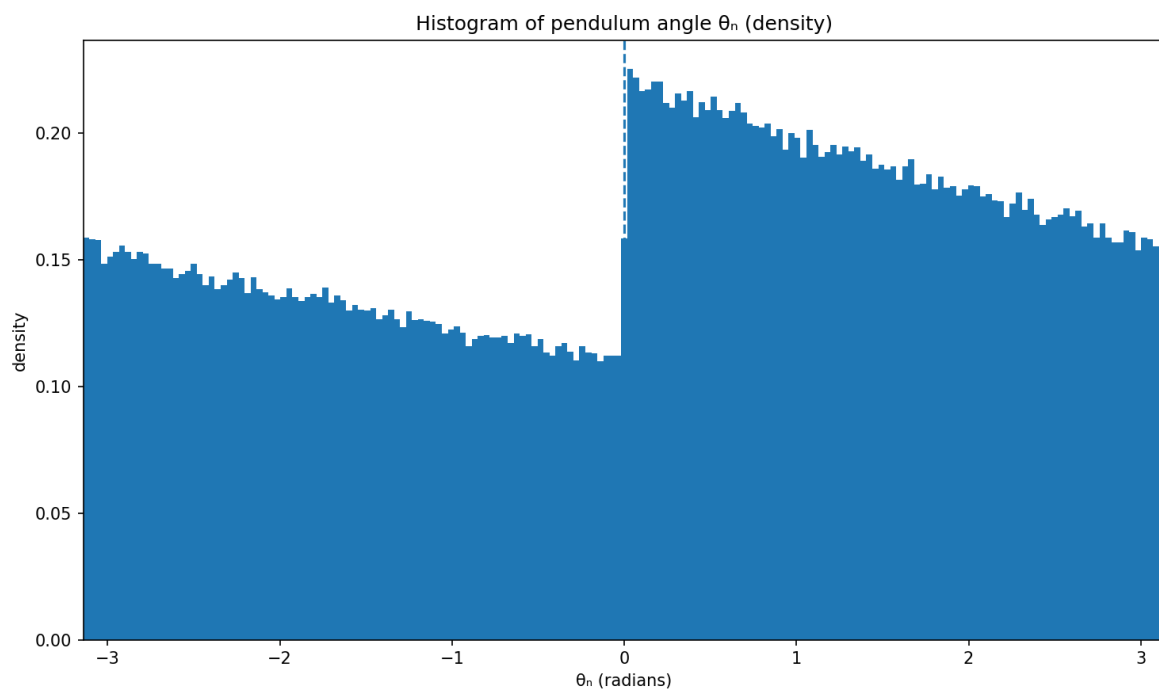
4. Key Statistical Results

4.1 Distribution of θ_n

- **Continuous distribution over $(-\pi, \pi]$**
- Moderate asymmetry (mean ≈ 0.18 rad)
- Approximately **41% of θ_n are negative**

Non-zero density near $\pm\pi$ ($\sim 0.8\%$)

θ_n effectively occupies the entire circle, with no structural gaps.



4.2 Distribution of α (factor p)

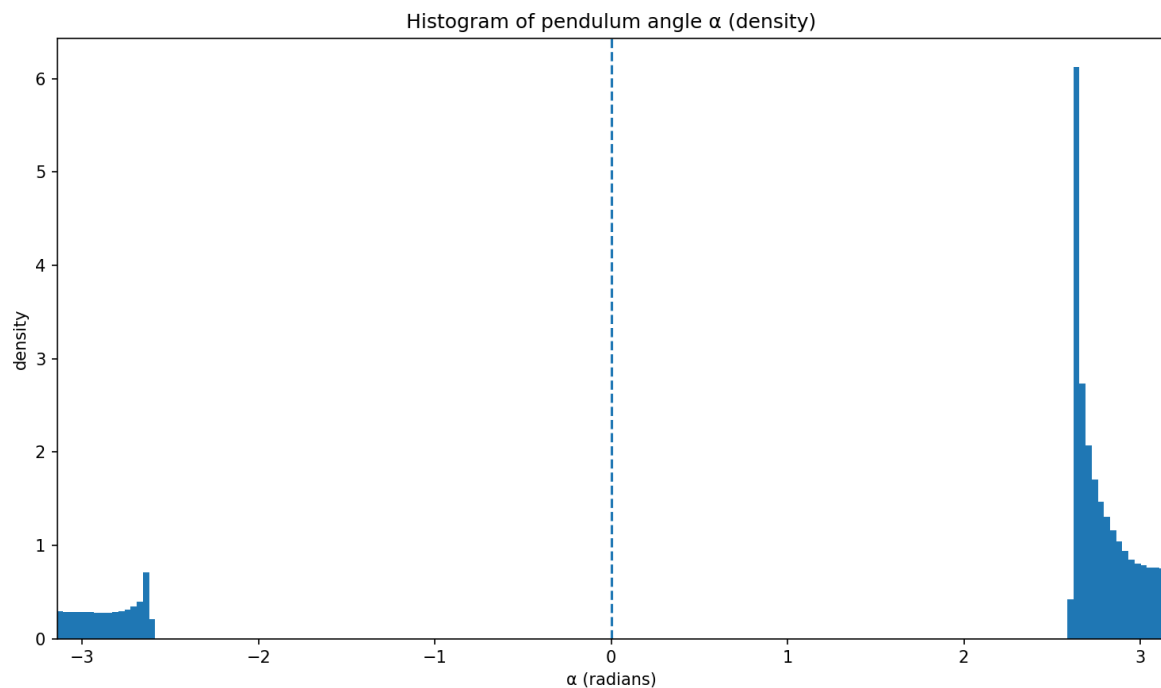
The distribution of α exhibits a **major structural asymmetry**:

- Massive concentration near $+\pi$
- Low population around $-\pi$
- Almost no intermediate values

Statistically :

- median ≈ 2.69 rad
- 90 % of values > 3.01 rad

α does not occupy a "lobe" in the broad sense, but rather two narrow neighborhoods of $\pm\pi$, with a strong asymmetry favoring $+\pi$.



4.3 Distribution of β (factor q)

The distribution of β is **strictly bounded**:

- Support $\approx [-\pi/2, +\pi/2]$
- Increasing density toward positive values
- No values near $\pm\pi$

β is not free on the circle, but confined to an angular band.

5. Conditioning of β by the Lobes of α

The circle is divided into π lobes as follows:

Positive lobe: $\alpha > +\pi/2$

Negative lobe: $\alpha < -\pi/2$

5.1 β | α in the positive lobe

- **Quasi-symmetric distribution**
- Peak around $\beta \approx 0$
- Large dispersion (std ≈ 0.76)

When α is in its dominant lobe, β is **weakly constrained**.

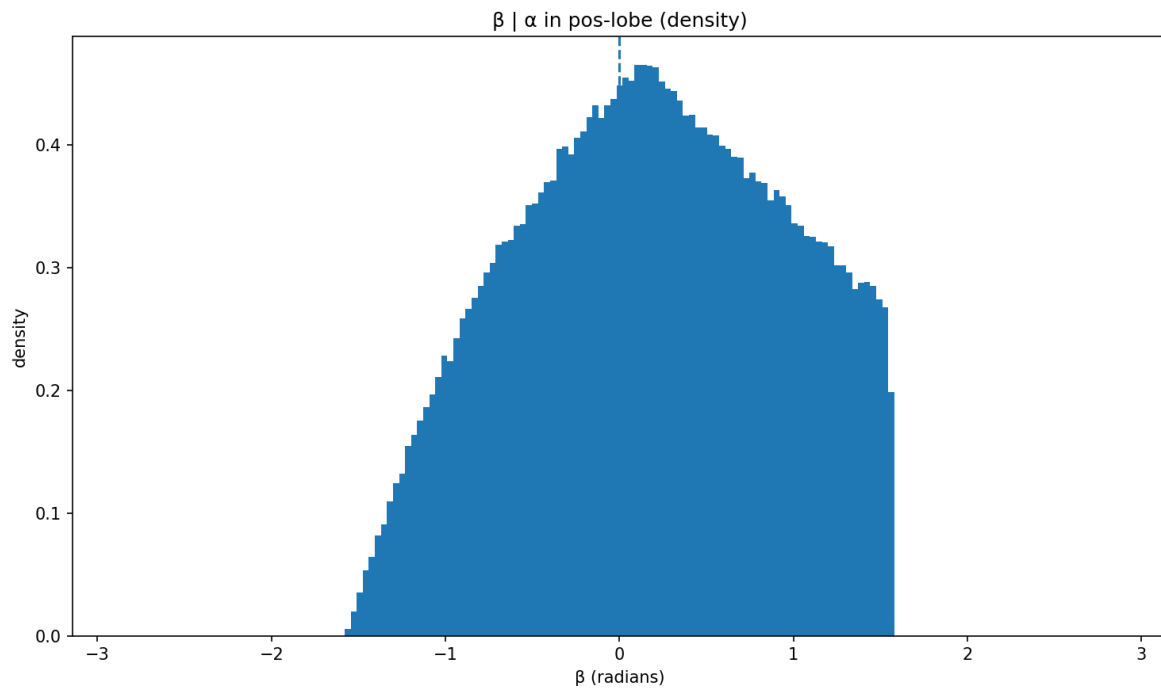
The figures clearly show that:

- **Negative α (lobe $-\pi$): β becomes strongly constrained**, positive, and close to $+\pi/2$
- **Positive α (lobe $+\pi$): β becomes broad again**, almost centered around 0

In other words :

- α acts as a **geometric switch**
- This is **not** a simple "complement" of α
- It is a **regime shift** in the admissible space of β

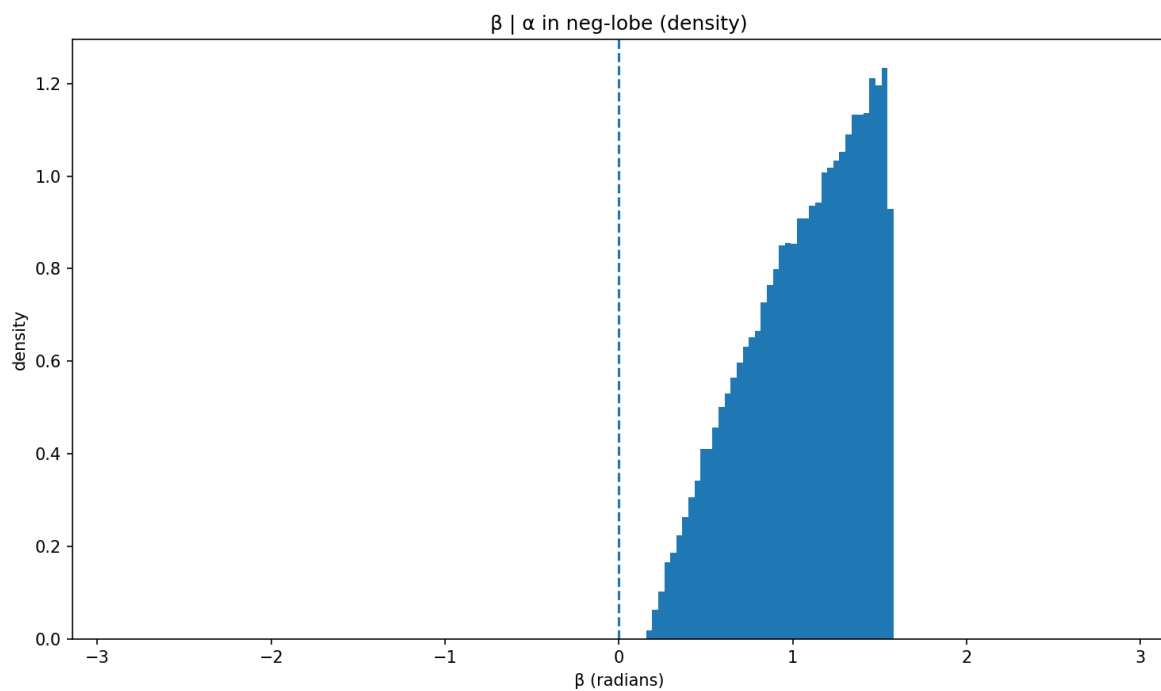
The transition of α from the neighborhood of $+\pi$ to $-\pi$ induces an abrupt change in the β regime, rather than a continuous variation.



5.2 $\beta \mid \alpha$ in the negative lobe

- **Strictly positive distribution**
- Strong concentration toward $\beta \approx \pi/2$
- Significantly reduced variance (std ≈ 0.34)

When α shifts to the opposite lobe, β tightens strongly toward the positive end.



6. Structural interpretation

These results show that:

1. $\alpha(p)$ plays the role of a **dominant phase variable**.

2. $\beta(q)$ is **geometrically conditioned** by the position of α .
3. The relationship $p-q$ is **neither independent nor symmetric**.
4. The angular information of p **strongly restricts the admissible space of q** .

This is a **global geometric effect**, not:

- A binning artifact
- A sampling bias
- A residue of random generation

7. Why This Is Not a Statistical Artifact

- Massive sample (500,000 triplets)
- Independent generation of angles
- Stability of histogram shapes
- Clear break between conditional lobes
- Significantly different variances

Random noise does **not** produce:

- Unilateral polarization of α
- Conditional confinement of β

8. Scope and exploitability

This structuring suggests:

An **exploitable angular reduction** of the search space for q

Non-trivial geometric information carried by p

A serious lead for:

Guiding factor search

Structural analysis of semiprimes

Multi-stage angular filters ($\alpha \rightarrow \beta$)

9. Next steps

- Replication on **KN127**
- Stability tests through subsampling
- Analysis of $\beta | \alpha$ with $\cos(\alpha)$ partitioning
- Projection into the $(\theta_n, \alpha, \beta)$ space
- Comparison with balanced/unbalanced semiprimes

Appendix A — Guided Reading of Angular Histograms (KN64)

A.1. General Conventions

The histograms presented in this study are derived from the file:

[pendulum_triplets_64_500k_with_beta.jsonl](#)

containing **500,000 triplets** (n , p , q) of 64-bit semiprimes.

The following conventions apply to **all** figures:

- Angles θ_n , α , and β are expressed in **radians**
- They are folded into the interval $(-\pi, +\pi]$
- The direction is **counterclockwise (trigonometric)**
- Density histograms represent a **relative occupation of the quadrant**, not an absolute probability

A.2. Histogram of θ_n — Quadrant of the Semiprime n

The angle θ_n is associated with the semiprime n and serves as the **global angular reference**.

Observations

- The distribution covers the **entire circle**
- Moderate asymmetry is visible
- Low density exists near $\pm\pi$ (~0.8%)

Correct Interpretation

- The quadrant of n is **globally occupied**
- There is no forbidden angular zone
- θ_n does **not** constitute a structuring filter by itself

θ_n provides a **reference**, not a constraint.

A.3. Histogram of α — Geometric Articulation $n \rightarrow p$

The angle α encodes the angular relationship between n and its prime factor p .

Key Observations

- The distribution is **strongly polarized**
- Two clusters appear near $\pm\pi$
- The lobe near $+\pi$ is **massively dominant**
- The central area of the quadrant is structurally sparsely populated

Correct Interpretation

- α is distributed **neither uniformly nor over a semicircle**
- It concentrates in **narrow antipodal neighborhoods**
- This polarization is **stable over 500,000 samples**

α acts as a **dominant articulation variable**, not a free angle.

A.4. Histogram of β — Geometric Articulation $p \rightarrow q$

The angle β describes the angular relationship between p and q , the second prime factor.

Key Observations

- β is strictly bounded in $[-\pi/2, +\pi/2]$
- No density appears near $\pm\pi$
- The distribution is **asymmetric**, with increasing density toward positive values

Correct Interpretation

- The quadrant associated with q is **geometrically restricted**
- β cannot occupy the entire circle
- This restriction is **structural**, not accidental

β is a **conditioned angular response**, not an independent variable.

A.5. Conditional Distribution $\beta \mid \alpha$ — Regime Shift

The circle is divided into two main lobes based on α :

- **Positive lobe:** $\alpha > +\pi/2$
- **Negative lobe:** $\alpha < -\pi/2$

Case 1 — α in the Positive Lobe (Majority)

- β exhibits a broad distribution
- Almost symmetric around 0
- High variance

When α is in its dominant lobe, **β remains weakly constrained**.

Case 2 — α in the Negative Lobe (Minority)

- β becomes **strictly positive**
- Strong concentration toward $+\pi/2$
- Significantly reduced variance

The shift of α to the opposite lobe induces a **brutal regime transition** for β .

A.6. What These Histograms Show (and Do Not Show)

They show:

- An **angular hierarchy** $\theta_n \rightarrow \alpha \rightarrow \beta$
- A fundamental asymmetry between p and q
- An exploitable conditional reduction of the angular space of q

They do not show:

- Direct factorization
- An explicit closed law
- A completed cryptographic algorithm

They reveal a **stable joint geometric structure**, absent from a naive random model.

Conclusion of the Appendix

The histograms should be read as **statistical projections of interacting quadrants**, not as independent distributions.

The polarization of α and the conditional confinement of β indicate that the internal structure of semiprimes possesses a **non-trivial angular organization**, which could be exploited in further analyses.